



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

End Semester Examination (Make-up)

July 2019

Max. Marks: 60

Class: S.E.

Course Code: CE43/IT44

Name of the Course: Operating Systems

Duration: 180 Min

Semester: IV

Branch: Computers/IT

Instruction:

- (1) All questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q No.	Question	Max. Mark	CO
Q.1(a)	Differentiate between Symmetric Multiprocessing and Asymmetric Multiprocessing. (Any 6 points)	06	CO1
	OR		
Q.1(a)	Differentiate between User mode and Kernel Mode.	06	CO1
Q.1(b)	What are the responsibilities of Operating System as Resource Manager?	06	CO1
Q.2	Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 1,805 and the previous request was at cylinder 2,150. The queue of pending requests, in FIFO order, is: 2020, 1220, 2290, 2810, 540, 350 Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? a. FCFS b. SSTF c. SCAN d. LOOK	12 [3M] [3M] [3M] [3M]	CO2
Q.3(a)	Illustrate and justify the solution for Dining Philosopher problem. (Pseudo-code expected)	06	CO3
	OR		

Q.3(a)	Evaluate the following the following code against all requirements of critical section solution. Consider only 2 processes wants to enter the critical section. Justify each requirement. Code: <pre> do { flag[i] = true; turn = j; while (flag[j] && turn == j); critical section flag[i] = false; remainder section } while (true); </pre>	06	CO3
Q.3(b)	Exemplify the Shortest Job First algorithm for Process scheduling. How exponential averaging burst prediction is used in Shortest Job First algorithm ?	06	CO2
Q.4(a)	Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)? Rank the algorithms in terms of how efficiently they use memory. (diagram is required) OR	06	CO4
Q.4(a)	Consider a system in which a program can be separated into two parts: code and data. The CPU knows whether it wants an instruction (instruction fetch) or data (data fetch or store). Therefore, two base-limit register pairs are provided: one for instructions and one for data. The instruction base-limit register pair is automatically read-only, so programs can be shared among different users. Analyze the scenario and discuss the advantages and disadvantages of this scheme.	06	CO4
Q.4(b)	Discuss the different indexing schemes used by Indexed File Allocation method with suitable diagram.	06	CO5
Q.5 (a)	Compare between the page replacement algorithms FIFO, Optimal and LRU.	06	CO4
Q.5(b)	Differentiate between different types of Page Tables.	06	CO4